Module 01: Introduction to Grafting

1.1 What is Grafting?

Grafting is a plant propagation technique where two different plant parts are joined together so they grow as a single, functional plant. This method is widely used in horticulture, agriculture, and commercial plant production to improve plant quality, productivity, and resilience.

Unlike growing plants from seeds, which introduces genetic variation, grafting allows growers to combine the best characteristics of two plants into one. This makes it a powerful tool for producing consistent, high-quality crops.

At its core, grafting involves two main components:

Scion – the upper part of the plant that will grow into stems, leaves, flowers, and fruits

Rootstock – the lower part that provides the root system and influences growth behavior

1.2 Key Components of a Graft

1. Scion (Upper Portion)

The scion is selected for its desirable traits such as:

High-quality fruit or flowers

Good yield

Specific variety characteristics

It becomes the visible, productive part of the plant after grafting.

2. Rootstock (Lower Portion)

The rootstock provides:

Water and nutrient absorption

Structural support

Resistance to soil-borne diseases

Control over plant size (e.g., dwarfing effect)

A well-chosen rootstock can significantly improve plant survival and performance.

1.3 The Science Behind Grafting

For grafting to succeed, the cambium layers of both the scion and rootstock must be properly aligned.

Cambium Layer

A thin layer of actively dividing cells located between the bark and wood

Responsible for forming new vascular tissues (xylem and phloem)

Essential for healing and union formation

When aligned correctly:

The plant forms callus tissue (a healing tissue)

The vascular systems reconnect

Water and nutrients begin to flow between both parts

This biological process allows the two separate plant parts to function as one.

1.4 Why Grafting is Used

Grafting is commonly used in commercial and home gardening for several reasons:

1. Combine Desirable Traits

High-quality fruit from the scion

Strong roots and disease resistance from the rootstock

2. Improve Plant Health

Rootstocks can resist pests, diseases, and poor soil conditions

3. Control Plant Size

Dwarfing rootstocks produce smaller, manageable trees

Ideal for limited spaces or intensive farming

4. Speed Up Production

Grafted plants often mature and produce fruit faster than seed-grown plants

5. Preserve Plant Varieties

Ensures genetic consistency (important for commercial crops)

1.5 Compatibility in Grafting

Not all plants can be grafted together. Successful grafting depends on genetic compatibility.

Plants within the same species or closely related species have higher success rates

Distantly related plants usually fail to form a proper union

Example:

Apple varieties can be grafted onto other apple rootstocks

Apple cannot be grafted onto unrelated plants like mango

Compatibility ensures proper healing and long-term survival.

1.6 Common Grafting Techniques (Overview)

While this course will explore techniques in later modules, here are some basic methods:

Whip/Splice Graft – used when scion and rootstock are similar in size

Cleft (Wedge) Graft – used when rootstock is thicker than the scion

Bud Grafting (Budding) – uses a single bud instead of a full stem

Each method is chosen based on plant size, species, and purpose.

1.7 Tools and Materials

Successful grafting requires proper tools:

- Sharp grafting knife (for clean cuts)
- Grafting tape or wrapping material (to hold the union)
- Wax or sealant (to prevent drying)

Important Principle:
Cuts must be clean and precise, and tools must be sanitized to prevent disease transmission.

1.8 Risks and Challenges

While grafting is effective, it comes with risks:

- Disease transmission if infected plant material is used
- Graft failure due to poor alignment or drying
- Incompatibility between scion and rootstock

Signs of failure include:

- Scion drying out or turning black
- Weak or broken union
- Rootstock producing unwanted shoots (suckers)

1.9 Healing and Aftercare

After grafting:

The union must be securely wrapped to maintain contact

Moisture loss should be minimized

Over time, the wrapping may need to be loosened to prevent girdling

Proper aftercare ensures the graft develops into a strong, functional plant.

1.10 Learning Outcomes of This Module

By the end of this module, you should be able to:

Define grafting and explain how it works

Identify the roles of scion and rootstock

Understand the importance of cambium alignment

Explain why grafting is used in agriculture and horticulture

Recognize basic grafting methods and tools